

Hadrien Tramoni

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EDUCATION

ENSSAT – École Nationale Supérieure des Sciences Appliquées et de Technologie <i>Engineering Degree (Diplôme d'Ingénieur) in Computer Science</i>	Lannion, France 2020 – 2024
IT University of Copenhagen <i>Exchange semester – Machine Learning, Computer Science</i>	Copenhagen, Denmark Aug 2022 – Jan 2023
Lycée Jean Dautet – Classes Préparatoires aux Grandes Écoles (CPGE) <i>Intensive undergraduate programme in Mathematics and Physics</i>	La Rochelle, France 2016 – 2020

PROFESSIONAL EXPERIENCE

Richemont International SA <i>Machine Learning Engineer</i>	Geneva, Switzerland Nov 2023 – Nov 2024
- Built a computer vision system that reads serial numbers from product images: designed the experiments, iterated on the models and reached over 90% accuracy on in-group items. - Applied LLMs (PaLM, Gemini) to extract structured data from unstructured documents; fine-tuned and evaluated the models on Vertex AI. - Took the prototypes to production as an API on GCP Cloud Run, with the infrastructure managed in Terraform. - Presented the results at Watches and Wonders 2024; managed a small team and coordinated priorities across several workstreams.	

PROJECTS

quant-modeling – Derivatives Pricing Library <i>Personal project, live demo at quant.tramonihadrien.com</i>	github.com/HadrienT/quant-modeling Jan 2026 – Present
- Wrote a C++20 pricing library (about 43k lines) that keeps instruments, models and engines separate: closed-form, binomial and trinomial trees, Crank-Nicolson PDE and Monte Carlo, for about 20 products across equity, rates, FX and commodities. - Calibrated volatility models on stored option chains: SVI per maturity, Dupire local volatility, Heston (COS pricing, Levenberg-Marquardt) and stochastic-local volatility by the particle method. - Implemented adjoint algorithmic differentiation (AAD) on an operator-overloading tape: the full vector of Greeks for roughly the cost of one extra valuation, instead of one bump per input. - Ported the Monte Carlo engine to CUDA on two Tesla V100 GPUs, bit-identical across GPU counts and within 1e-12 of the CPU; a daily-monitored barrier option under local volatility reaches a 1e-4 relative standard error in 12.7 s, against 27 min on 8 CPU threads. - Added a payoff scripting language (lexer, parser, AST) so that one generic Monte Carlo engine prices any scripted payoff, and an exposure engine computing the CVA of a collateralised netting set. - Tested with about 70 GoogleTest suites that favour properties (put-call parity, in + out = vanilla, measured convergence order) and with ASan/UBSan builds; exposed through pybind11, FastAPI and a React front end, self-hosted with Docker and GitHub Actions CI. - Built alongside the reading listed below, with an AI coding assistant (Claude Code).	

INDEPENDENT STUDY IN QUANTITATIVE FINANCE

<i>Self-directed, full time</i>	Nov 2024 – Present
- Derivatives and stochastic calculus: Hull, <i>Options, Futures, and Other Derivatives</i>; Shreve, <i>Stochastic Calculus for Finance I and II</i>. - Volatility: Gatheral, <i>The Volatility Surface</i>; Bergomi, <i>Stochastic Volatility Modeling</i>; Bennett, <i>Trading Volatility</i>. - Numerical methods: Glasserman, <i>Monte Carlo Methods in Financial Engineering</i>; Savine, <i>Modern Computational Finance: AAD and Parallel Simulations</i>; Hwu, Kirk and El Hajj, <i>Programming Massively Parallel Processors</i>; Tsay, <i>Analysis of Financial Time Series</i>.	

SKILLS

Programming: C++ (C++20), Python, CUDA, C, SQL, TypeScript
Quantitative finance: Monte Carlo and variance reduction (antithetic and control variates, Sobol, Brownian bridge), finite-difference PDE, trees, local and stochastic volatility, short-rate models (Vasicek, Hull-White), calibration, Greeks
Libraries and tools: CMake, GoogleTest, pybind11, Eigen, NumPy, pandas, PyTorch, scikit-learn, FastAPI, PostgreSQL
Infrastructure: Linux, Git, Docker, GitHub Actions (CI/CD), GCP (Cloud Run, BigQuery), Terraform
Languages: French (native), English (C1), German (B2)